

Figure 2: Accessing a local VM using the libvirt API

list of processors supporting virtualisation.

Using KVM, you can run multiple virtual machines, where each VM is a separate Linux process, and is scheduled by the Linux scheduler just like any other process. Each virtual machine can run an unmodified Linux (or Windows) image. Each of these virtual machines have private virtualised hardware—viz., a network card, disk, graphics adapter, etc.

Note that any normal Linux process has two modes: the kernel mode and user mode. In kernel mode, privileged instructions are executed within the kernel. In user mode, non-privileged instructions are executed in user space. KVM adds a third mode called the guest mode, and certain instructions of the guest OS are executed in this mode.

For more details on how to work with KVM, please refer to: [www.linuxforu.com/how-to/kvm-virtualisation-the-linux-way](http://www.linuxforu.com/how-to/kvm-virtualisation-the-linux-way). Figure 1 gives an architectural overview of KVM.

## JeOS

Now that we're done with an overview on KVM, let's get familiar with another term—Ubuntu JeOS (just-enough operating system).

Quoting the Ubuntu Documentation [[help.ubuntu.com/community/JeOS](http://help.ubuntu.com/community/JeOS)]: "Ubuntu JeOS (pronounced 'Juice') is an efficient variant of the Ubuntu Server OS, configured specifically for virtual appliances. It comes with a tuned kernel that only contains the base elements needed to run within a virtualised environment. Ubuntu JeOS excludes unnecessary drivers, and retains only the minimal required packages."

You can create an Ubuntu JeOS-based VM using a program called `ubuntu-vm-builder`—described in more detail in the next section.

## Installing KVM

The first step is to install KVM and associated programs:

```
$ sudo aptitude install kvm libvirt-bin \
ubuntu-vm-builder bridge-utils
```

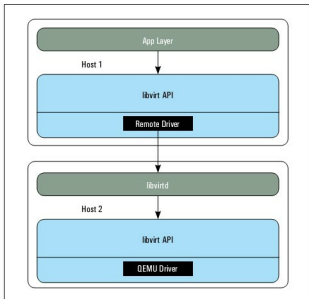


Figure 3: Accessing a remote VM using the libvirt API

The following are the package descriptions:

- `kvm` is the KVM driver.
- `libvirt-bin` provides `libvirtd` which is needed to administer QEMU and KVM instances using `libvirt`.
- `ubuntu-vm-builder` is a powerful command-line tool for building JeOS virtual machines.
- `bridge-utils` provides utilities to create a bridge from the virtual machines to the physical network interface (e.g., an Ethernet card).

A default virtual bridge interface `virbr0` is created automatically when installing KVM. In case bridged-networking is required, a `br0` bridge will need to be created in `/etc/network/interfaces`.

The next step is to create a JeOS virtual machine using `ubuntu-vm-builder` as follows:

```
$ sudo ubuntu-vm-builder kvm karmic --arch 'amd64' \
--mem '1024' --rootsize '10240' --swapspace '2048' \
--kernel-flavour 'server' --hostname 'yb-euc1' \
--mirror 'http://archive.ubuntu.com/ubuntu/' \
--components 'main,universe' --name 'admin' \
--user 'admin-server' --pass 'welcome123' \
--ip '192.168.0.20' --libvirt qemu:///system
```

Here:

- `vm` is the target virtualisation software—in this case, KVM.
- `suite` is the specified Ubuntu suite (Karmic).
- `arch` is the target architecture (amd64, i386).
- `mem` is the memory in MB, of the RAM (1024 MB).
- `rootsize` is the disk space for the root filesystem (10240 MB = 10GB).
- `swapspace` is the swap space (2 GB).
- `kernel-flavour` is the kernel image flavour to install (Ubuntu server).